

Tissue-Specific Hit Calling

Kexin Dong | B-ALL In Vivo Base Editing Screen

Guide-level filtering (applied before hit calling)

Filter	Focused screens	OG screens
Input_median	≥ 2	≥ 2
sensor_reads_{cond}	≥ 100 (any tissue)	≥ 100 (any tissue)
target_base_edit_perc_{cond}	$\geq 20\%$ (any tissue)	$\geq 20\%$ (any tissue)

Hit category definitions

All categories use **LFC median** effect, **Fisher's combined p-value** FDR method, with thresholds **FDR < 0.1** and **|LFC| > 1**.

"**Not a strong enricher in tissue X**" = NOT (FDR < 0.1 AND LFC ≥ 1)

This captures three allowed states: (1) not significant, (2) significant but weak enricher ($0 < \text{LFC} < 1$), or (3) significant depleter (LFC < 0).

1. In Vitro-Specific

- FDR < 0.1 AND LFC > 1 at **d15**
- Not a strong enricher in **all** in vivo tissues (spleen, BM, $\pm$ meninges)

2. In Vivo-Specific

- Not a strong enricher at **d15**
- FDR < 0.1 AND LFC > 1 in **at least one** in vivo tissue

3. Spleen-Specific

- FDR < 0.1 AND LFC > 1 in **spleen**
- Not a strong enricher in **d15** and **BM**

4. BM-Specific

- FDR < 0.1 AND LFC > 1 in **BM**
- Not a strong enricher in **d15** and **spleen**

Screens compared

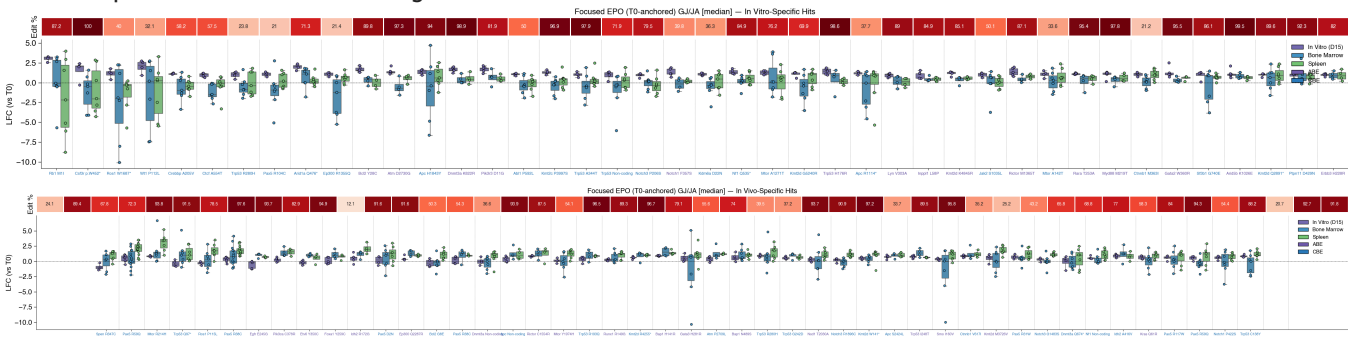
Label	ABE key	CBE key	Conditions	Notes
GJ/JA T0	ABE_EPO_T0	CBE_EPO_T0	d15, spleen, BM	Focused, T0-anchored
GJ/JA BC	ABE_EPO_BC	CBE_EPO_BC	d15, spleen, BM	Focused, BC-anchored
FSR T0	ABE_EPO_T0_FSR	CBE_EPO_T0_FSR	d15, spleen, BM	FSR replicates
FSR BC	ABE_EPO_BC_FSR	CBE_EPO_BC_FSR	d15, spleen, BM	FSR replicates
OG	ABE_OG	CBE_OG	d15, spleen, BM, meninges	OG library

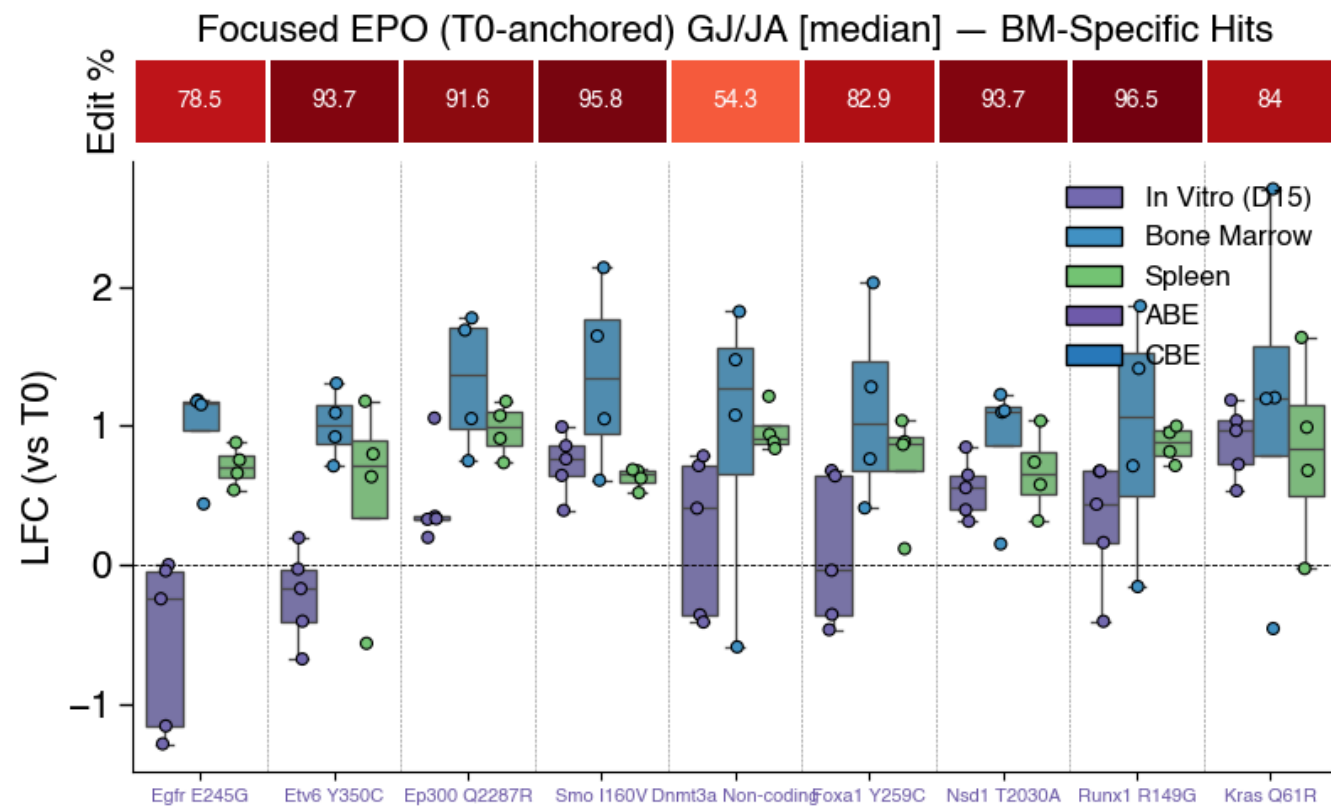
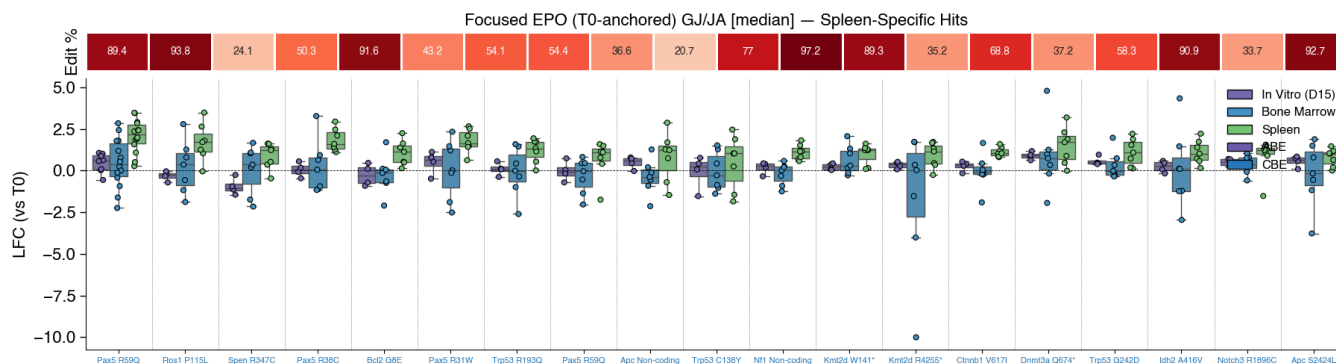
Loaded:

ABE_OG: 1070 guides
CBE_OG: 13242 guides
ABE_EPO_T0: 376 guides
CBE_EPO_T0: 1257 guides
ABE_EPO_BC: 376 guides
CBE_EPO_BC: 1257 guides
ABE_BC_T0: 376 guides
CBE_BC_T0: 1257 guides
ABE_EPO_T0_FSR: 376 guides
CBE_EPO_T0_FSR: 1257 guides
ABE_EPO_BC_FSR: 376 guides
CBE_EPO_BC_FSR: 1257 guides
ABE_BC_T0_FSR: 376 guides
CBE_BC_T0_FSR: 1257 guides

=== Focused EPO (T0-anchored) GJ/JA | effect=median ===

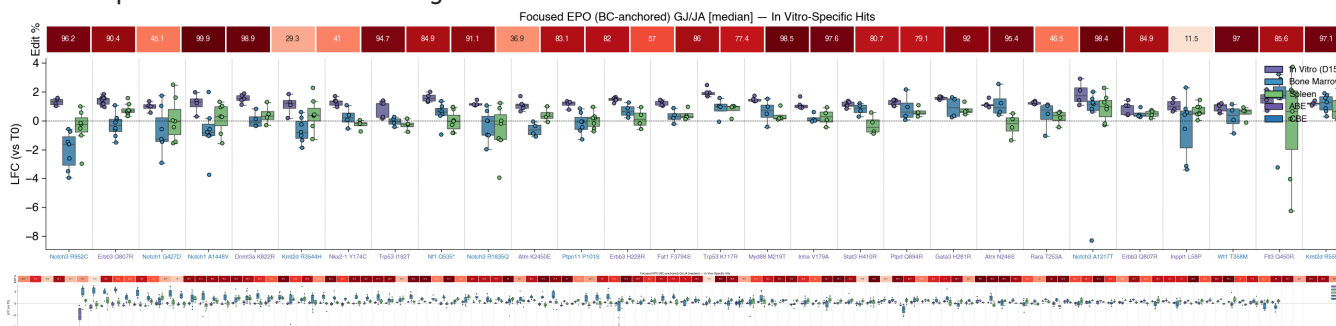
In vitro-specific : 42 guides
In vivo-specific : 48 guides
Spleen-specific : 31 guides
BM-specific : 9 guides

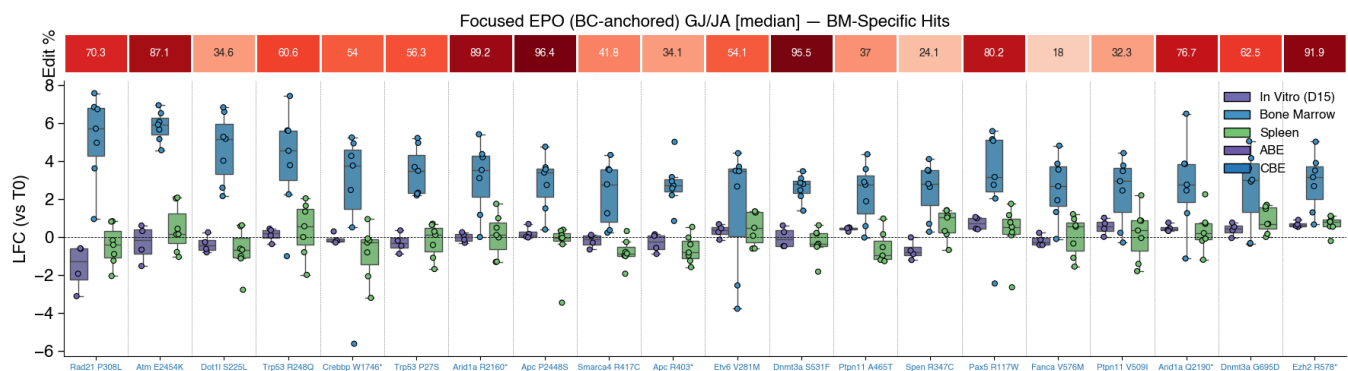
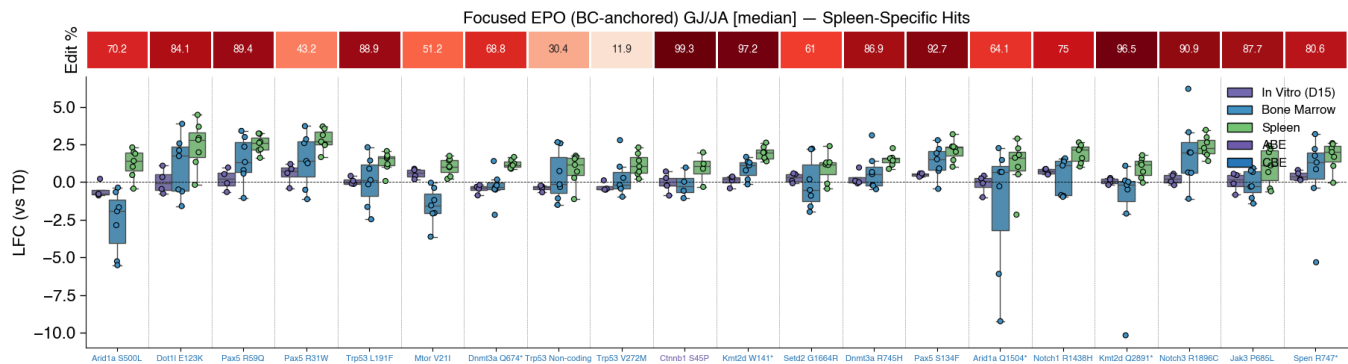




=== Focused EPO (BC-anchored) GJ/JA | effect=median ===

In vitro-specific : 29 guides
 In vivo-specific : 113 guides
 Spleen-specific : 50 guides
 BM-specific : 49 guides





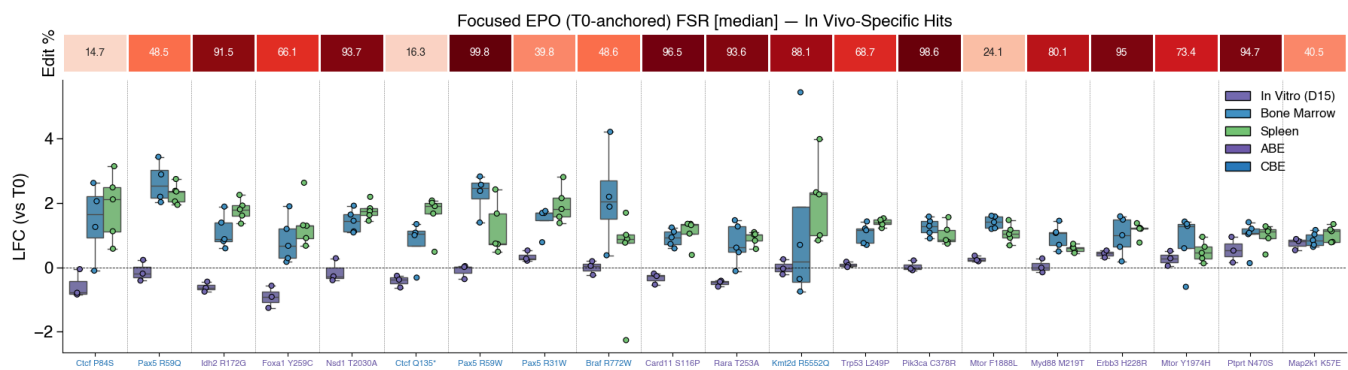
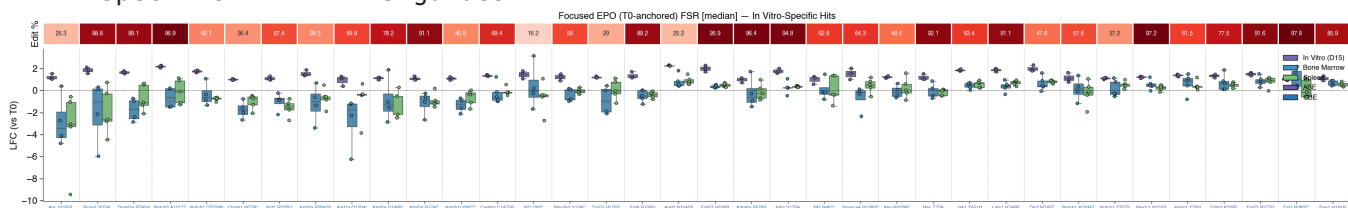
=== Focused EPO (T0-anchored) FSR | effect=median ===

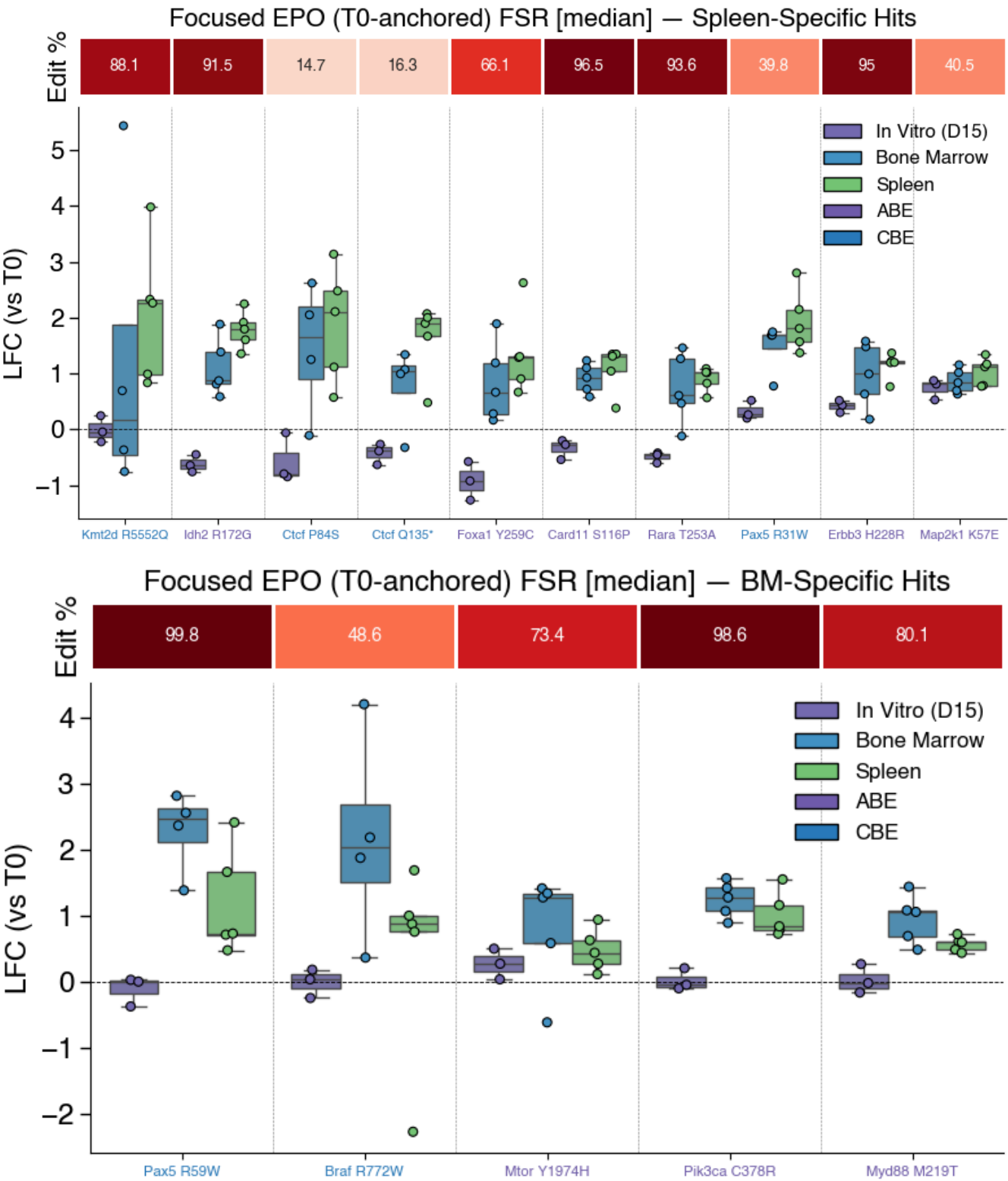
In vitro-specific : 36 guides

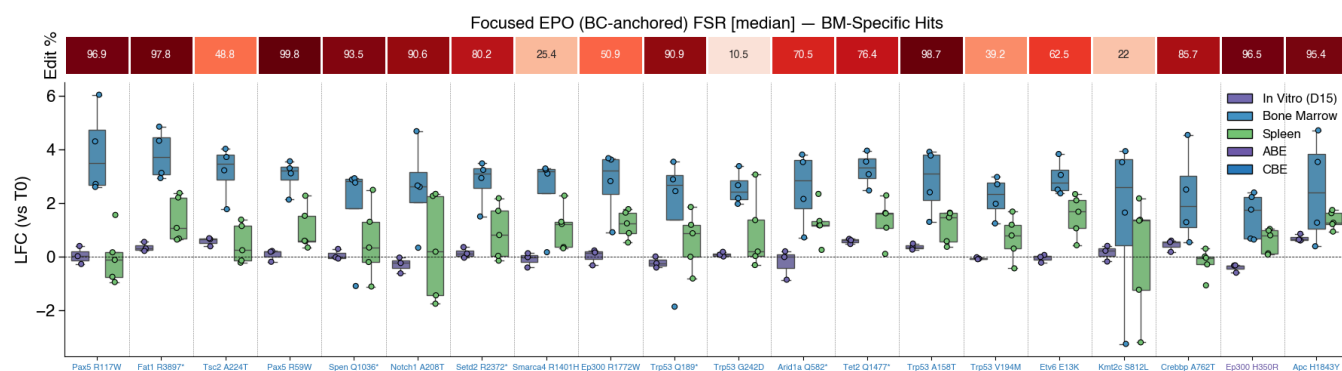
In vivo-specific : 20 guides

Spleen-specific : 10 guides

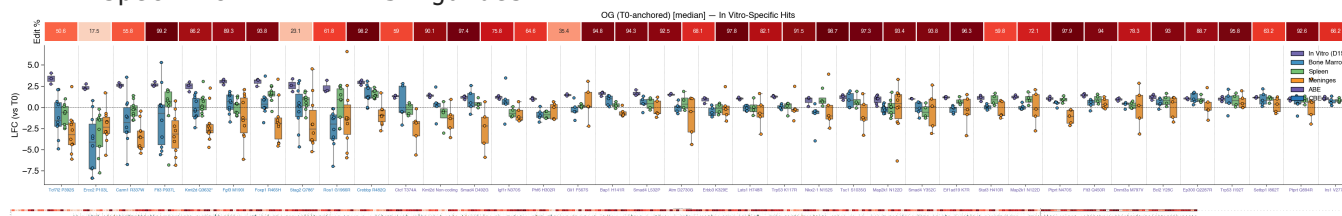
BM-specific : 5 guides

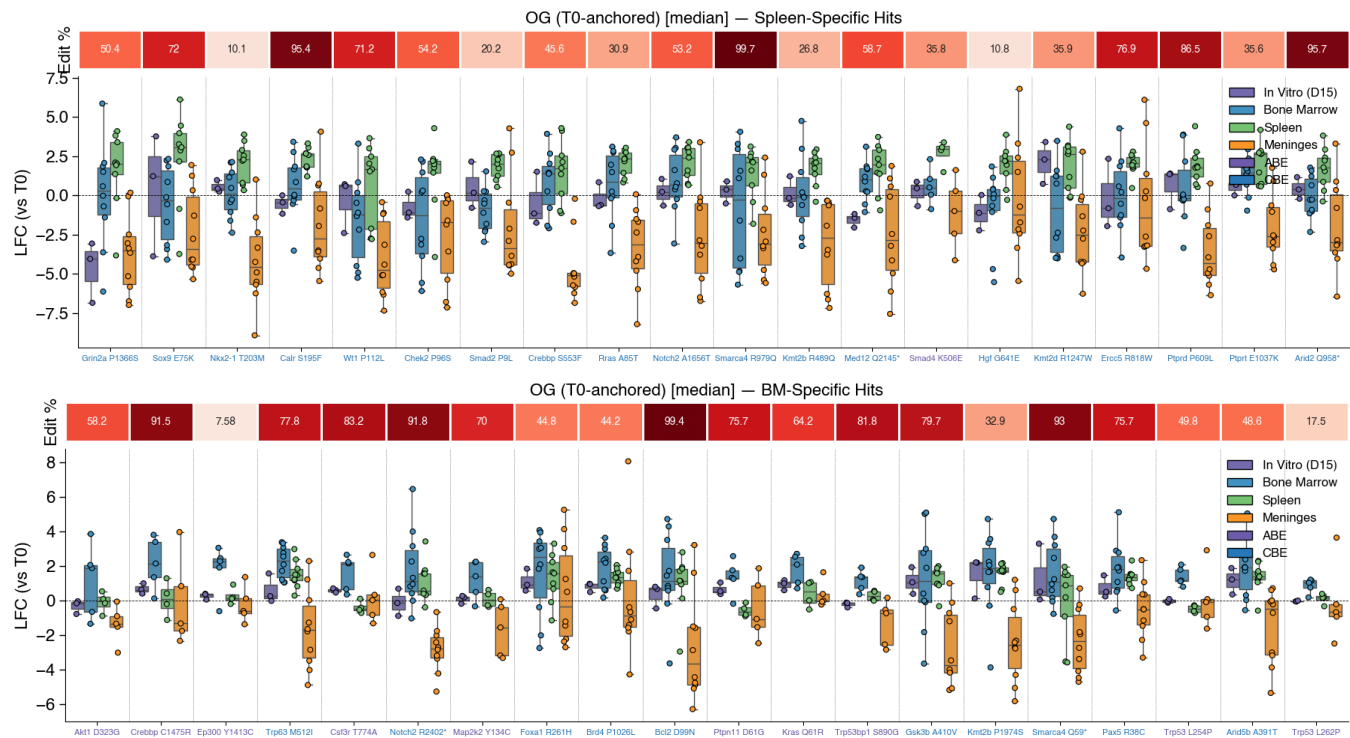






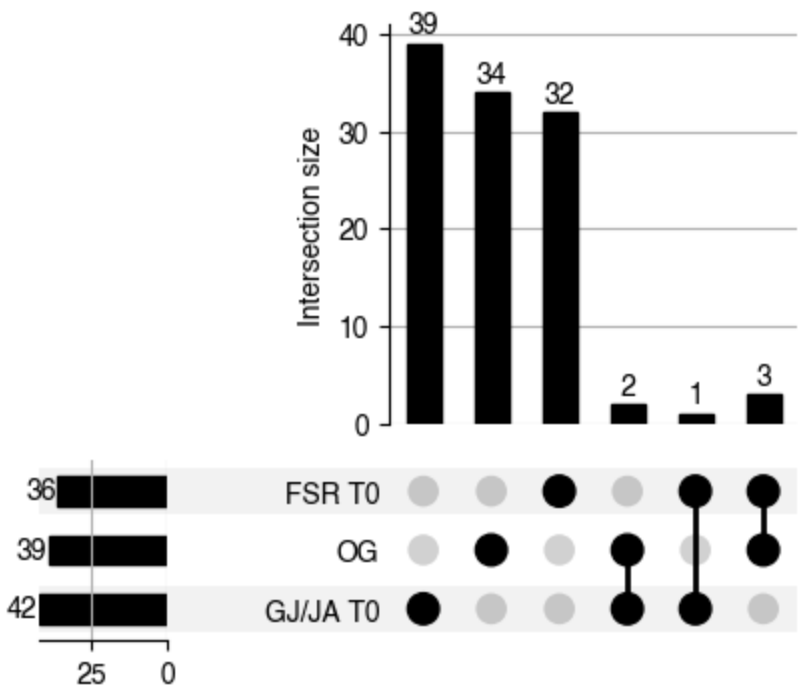
In vitro-specific	: 39 guides
In vivo-specific	: 412 guides
Spleen-specific	: 271 guides
BM-specific	: 34 guides

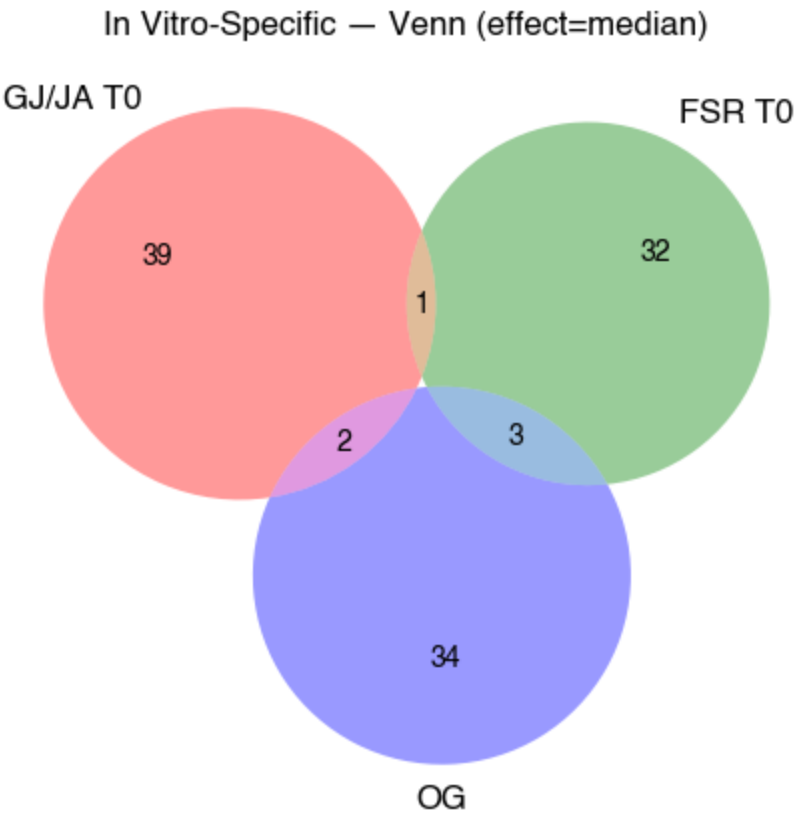




In Vitro-Specific
GJ/JA T0: 42 guides
FSR T0: 36 guides
OG: 39 guides

In Vitro-Specific — UpSet (effect=median)

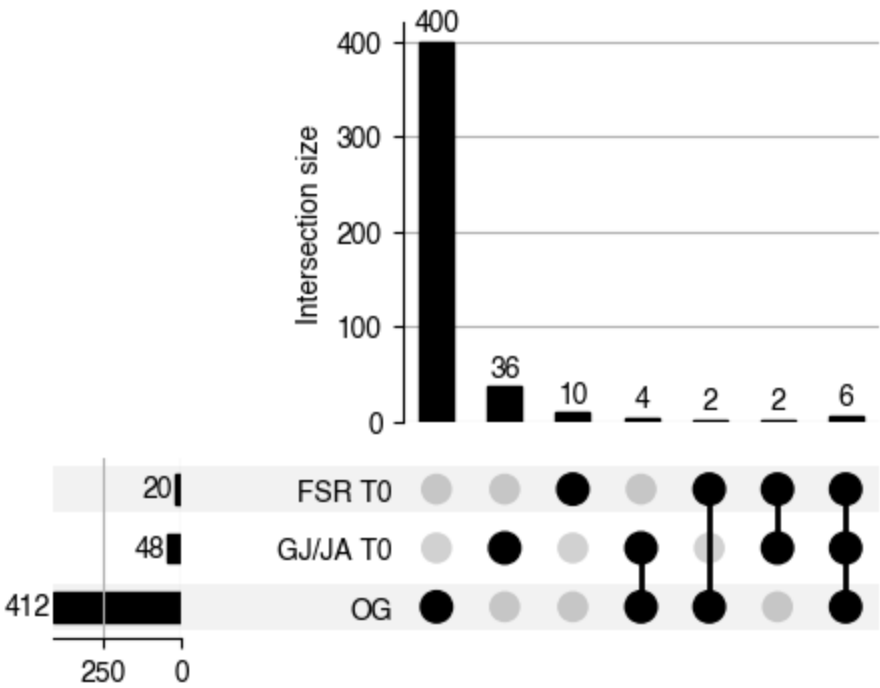




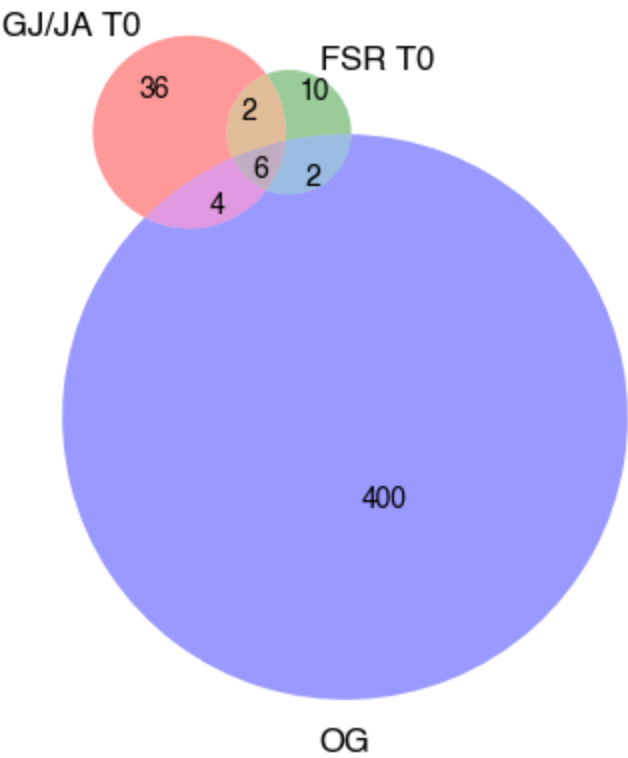
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In Vivo-Specific
GJ/JA T0: 48 guides
FSR T0: 20 guides
OG: 412 guides

In Vivo-Specific — UpSet (effect=median)



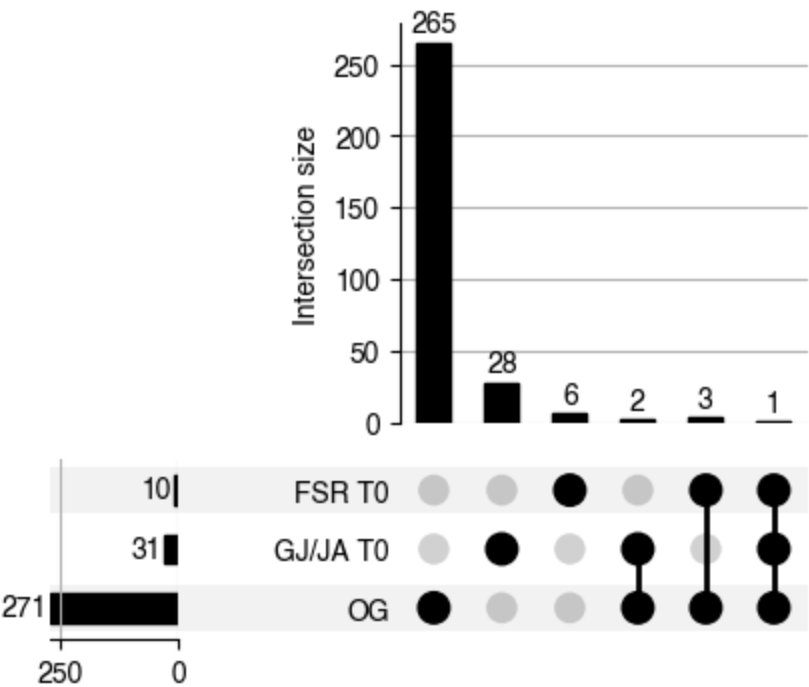
In Vivo-Specific — Venn (effect=median)



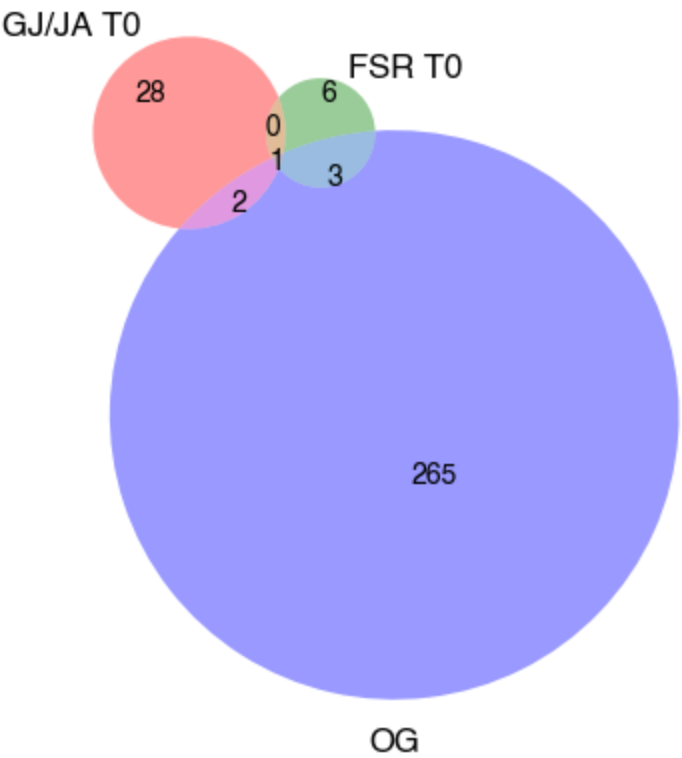
Spleen-Specific

GJ/JA T0: 31 guides
FSR T0: 10 guides
OG: 271 guides

Spleen-Specific — UpSet (effect=median)



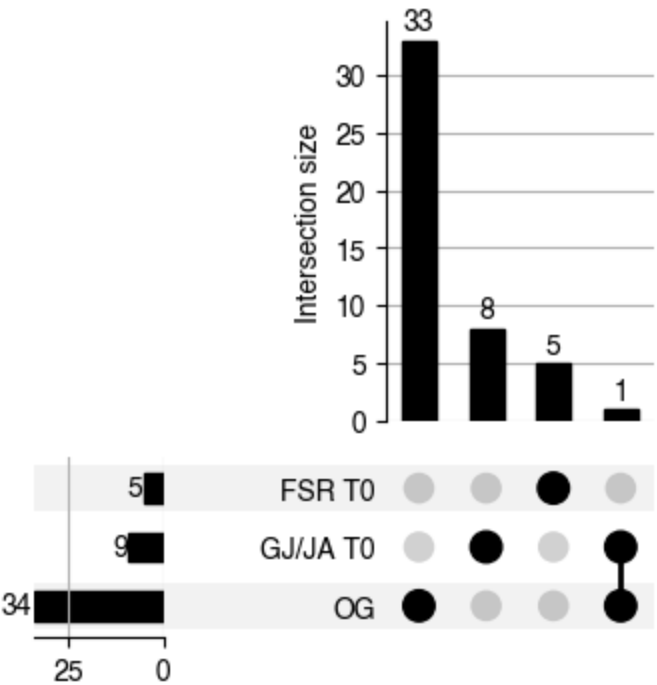
Spleen-Specific — Venn (effect=median)

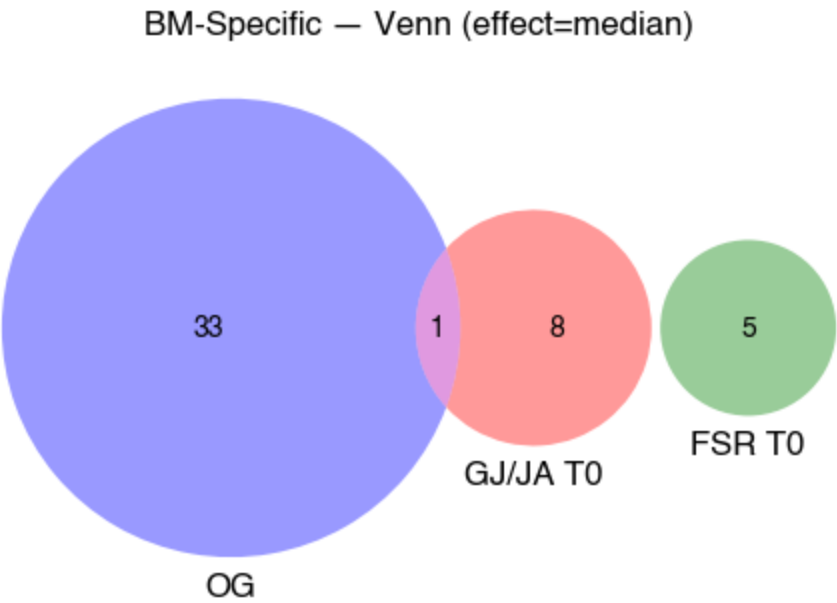


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BM-Specific
GJ/JA T0: 9 guides
FSR T0: 5 guides
OG: 34 guides

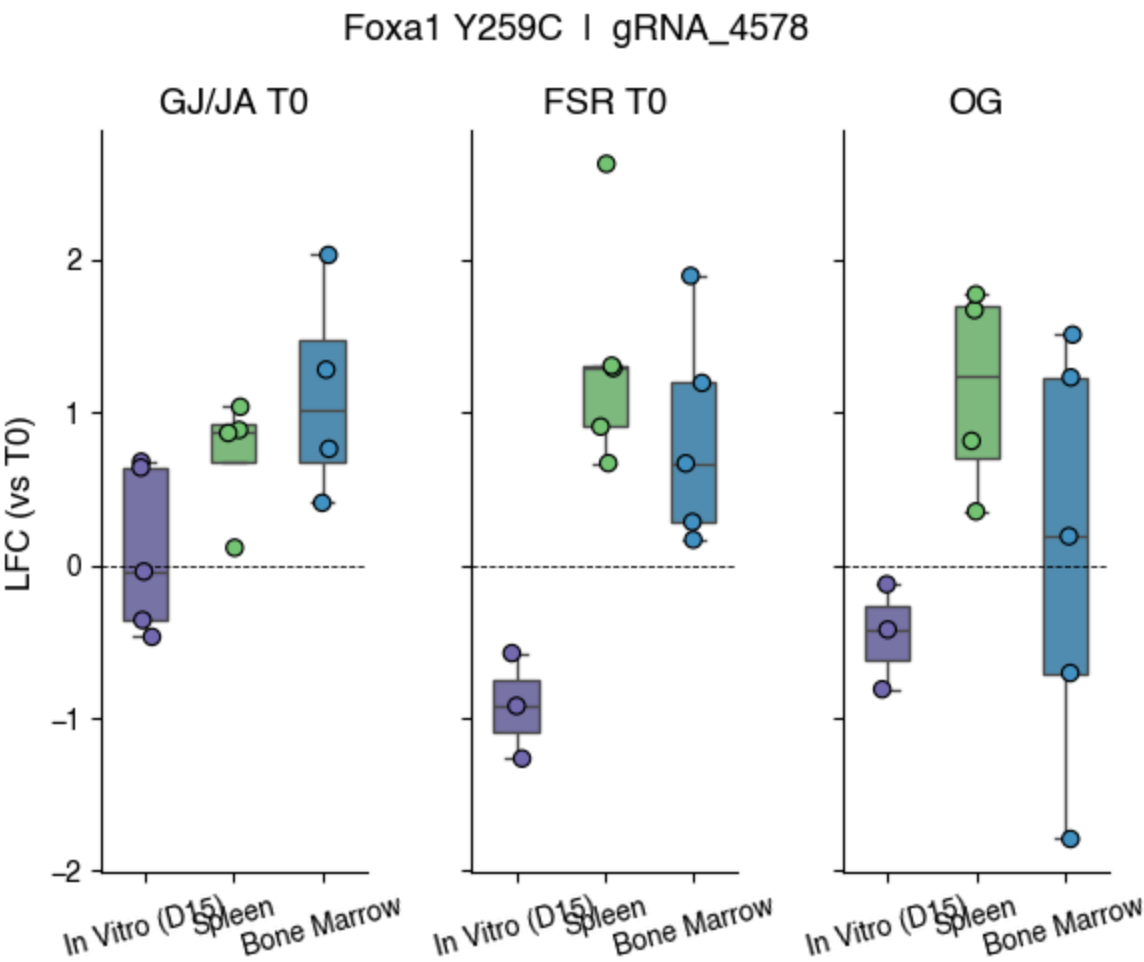
BM-Specific — UpSet (effect=median)



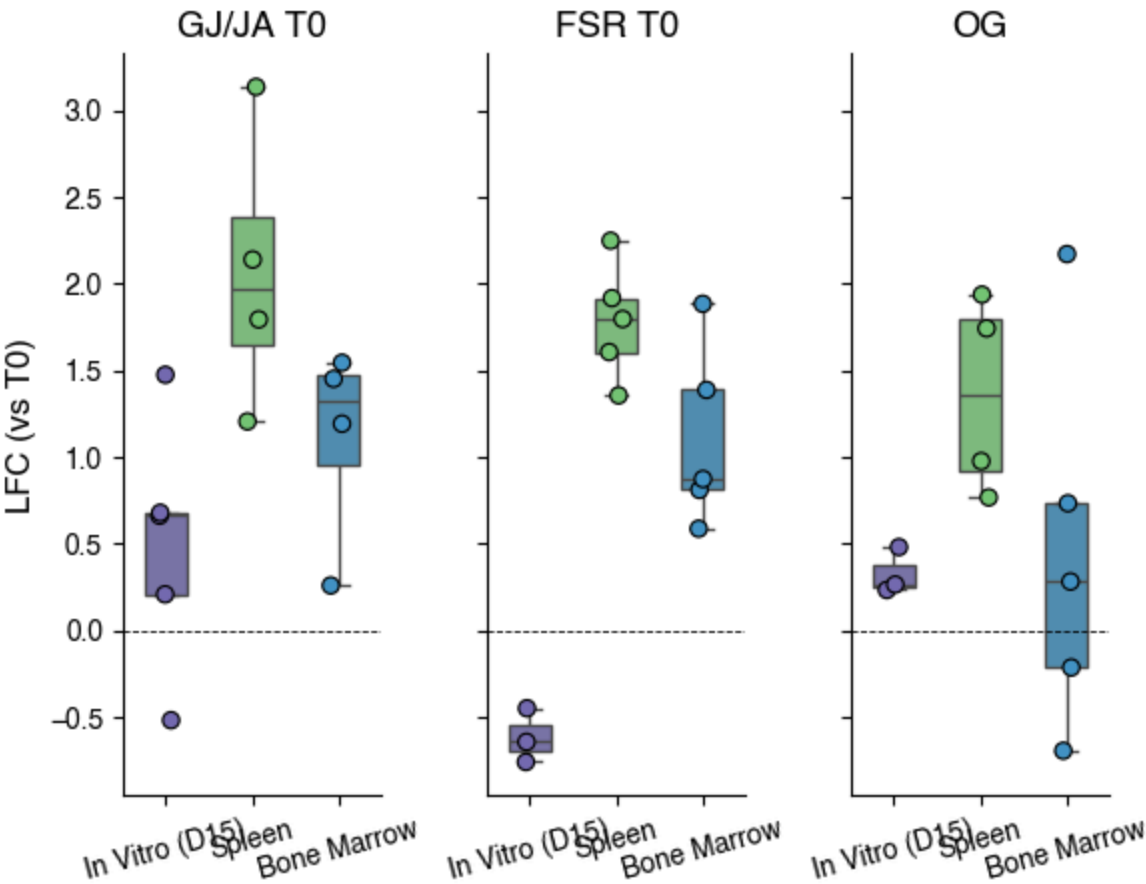


plot share in vivo specific hits

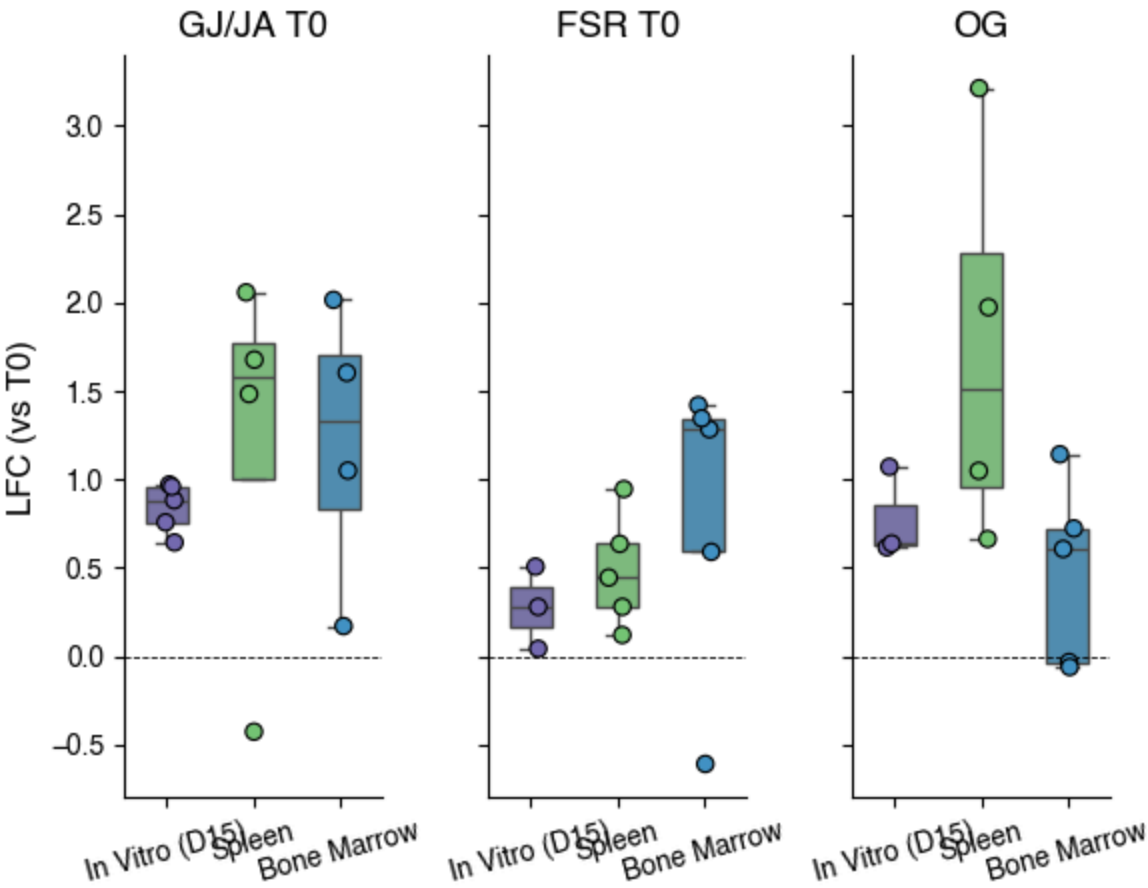
Triple overlap: 6 guides



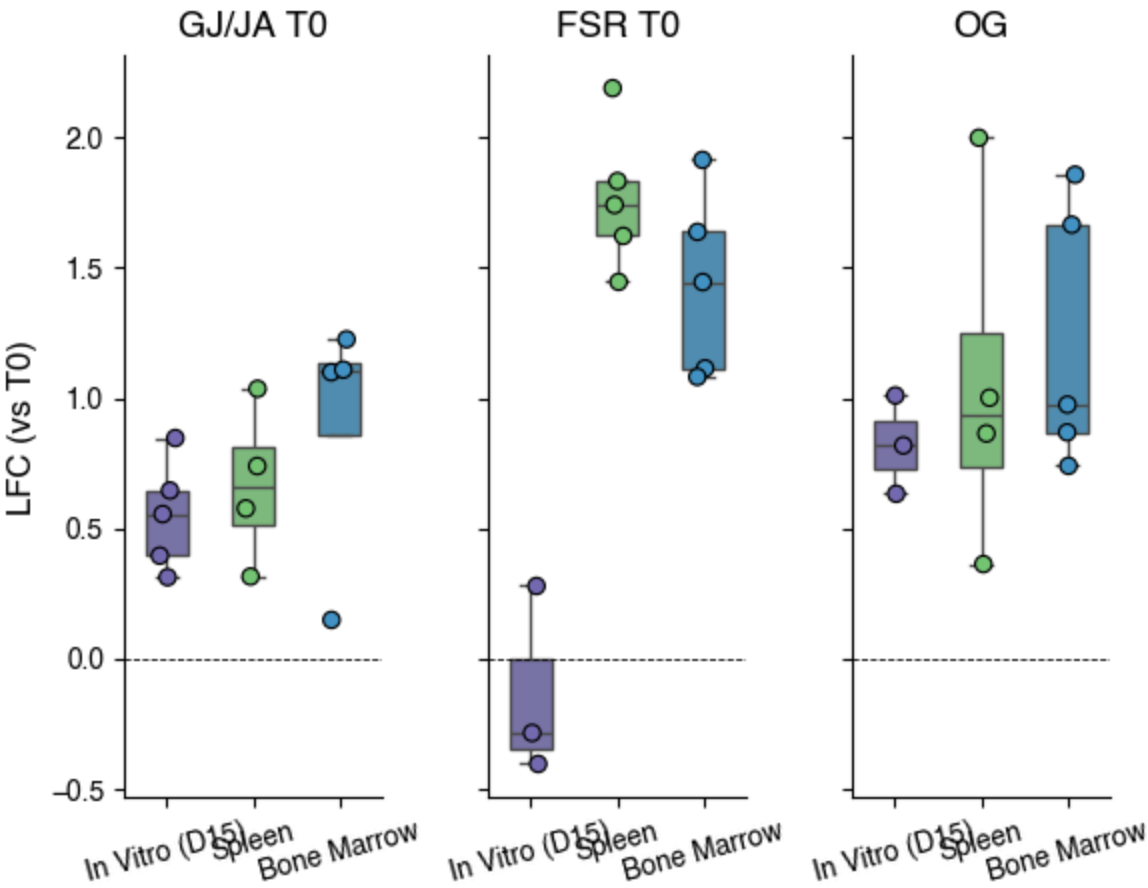
Idh2 R172G | gRNA_5194



Mtor Y1974H | gRNA_7393



Nsd1 T2030A | gRNA_8228



Pax5 R31W | gRNA_8516

